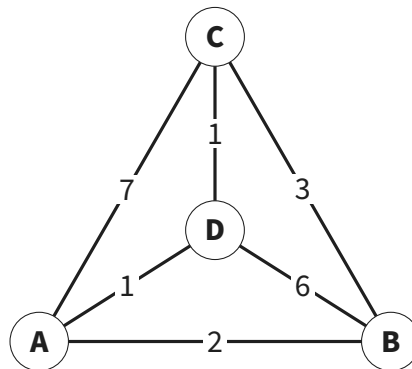


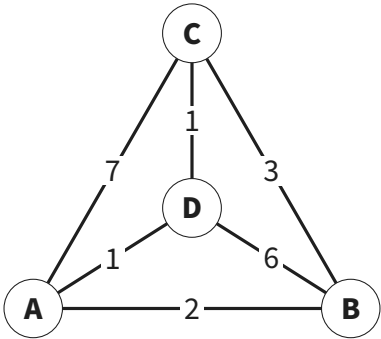
Problem 1: Start at A, travel every road at least once, and return to A. Numbers are road lengths; every repeat costs that length again. Find two complete routes. Record each visited dot and add the traveled lengths. Circle your shorter route.



Trial	Route	Length
1		
2		

Problem 2: Add the lengths of the six original roads. In one of your routes, mark the first trip along each road. List only the extra trips and their total cost. Even a long road must be traveled once.

Problem 3: Find the cheapest connection between each pair of dots. A connection may follow several roads. Compare the direct road with a route through another dot; record the cheapest route and its cost.

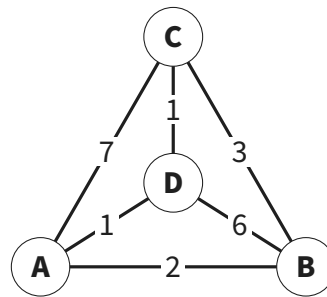
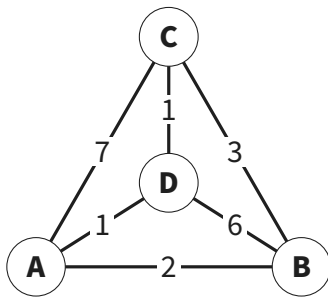


Pair	Direct cost	Cheapest connection	Cost
AB	2		
AC	7		
AD	1		
BC	3		
BD	6		
CD	1		

Problem 4: The four dots all have odd degree. Pair them in each way below and add the two cheapest connection costs. Compare the extra costs. Circle the best pairing.

Pairing	Total extra cost
AB + CD	
AC + BD	
AD + BC	

Problem 5: On the left map mark extra trips for your best pairing. On the right mark extra trips for $AD + BC$. Find a closed route on each map using every original and extra road copy exactly once. Record both routes and total lengths.



Problem 6: A proposed rule is “Choose a cheapest pair first, then connect the remaining two dots.” Choose AD first and calculate the result. Choose CD first and calculate the result. Record whether the rule always finds the best total in this example.

Problem 7: In your best route, erase one required copy of each original road from its crossing list. Draw just the extra road copies below. Circle their odd-degree endpoints and match them with one of the pairings in Problem 4.